

MALAKA PERERA

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Visa Status: 190 Permanent Resident | Target Location: Melbourne, VIC, Australia

PROFESSIONAL SUMMARY

Accomplished Textile R&D Leader and textile Material Engineer with over 10 years of experience delivering commercially successful, sustainable, and high-performance material innovations for global apparel brands. Expert in performance fabric development, cost-efficient manufacturing models, and material quality optimization. Proven track record of leading cross-functional teams, driving open innovation with global supply chains, and translating material science into scalable, retail-ready garments.

KEY SKILLS

- **Advanced Textile Engineering:** Smart Textiles, Advanced Flatbed Knitting (Stoll & Shima Seiki), 3D Knitting, Functional textile composites, Zonal Fabric properties, Wearable Sensors & Actuators Integration
- **Research & Development:** Material White-Space & Trend Analysis, Natural fiber engineered solutions, Wearable Electronics, Pilling Performance Optimization, Moisture Management, Passive Cooling & Heating, Textile material opaqueness and translucency, Hydromorphic adaptive textile material.
- **Specialized Industry Software:** M1 Plus (Stoll Programming), Shima Apex (Shima Seiki Whole-Garment Programming)
- **Project & Open Innovation:** Cross-Functional Team Leadership, Open Innovation Methodologies, Cost Engineering, Product Lifecycle Management (PLM), Stakeholder Management & Technical Consulting
- **Computer & Design Skills:** Microsoft Office (Word, Excel, PowerPoint, Project), Adobe Creative Suite (Photoshop, After Effects, Premiere Pro, Illustrator)
- **Languages:** English (Fluent), Sinhala (First Language)

WORK EXPERIENCE

Deputy General Manager – Technology Entrepreneur & Textile Engineer

TWINERY, INNOVATION BY MAS | Colombo, Sri Lanka | Full-Time

April 2025 – Present

- Led material white-space and trend analysis at the Material Center of Excellence to drive strategic wide MAS Holdings innovation initiatives.
- Defined and executed a comprehensive material excellence roadmap aligned with future-focused advanced manufacturing strategies.
- Conducted structured R&D on adaptive textile materials (including hydromorphic, shape-memory polymers, and phase-change materials), analyzing smart material behaviors in knit structure and use cases for everyday clothing.
- Engineered lightshade, squat-proof leggings materials optimized for consistent opacity and structural integrity under 60% stretch.
- Developed advanced soft-good textile covering systems for robotic applications for a leading global robotics brand.

Manager – R&D in Flatbed Knitting Technology

TWINERY, INNOVATION BY MAS | Colombo, Sri Lanka | Full-Time

November 2021 – April 2025

- Acted as Project Lead for onEKnit (Next S-curve for Flatbed Knitting technology), coordinating cross-functional engineering teams to implement decentralized, customized manufacturing solutions.
- Managed comprehensive cost engineering for novel business models, successfully reducing material costs by 28% (product, process, and transfer price-related cost optimization).

- Led the development of the textile component of an engineered knitted active pneumatic compression medical sleeve.
- Engineered active vibration therapy knitted sleeves with seamlessly integrated vibration motors.
- Enabled a sustainable business model for on-demand customization, rapid prototyping, and near-shoring using flatbed knitting technology.
- Developed innovative yarn solutions utilizing Siro compact yarn spinning to resolve pilling issues in regenerated fibers, successfully improving pilling performance from a rating of 2 to 3.5.
- Developed a 100% recyclable, mono-material bra cup solution utilizing advanced 3D knitting technology to replace traditional polyurethane cups.
- Provided specialized knit-related technical consultation to MAS Holdings' network of manufacturing facilities.
- Invented and filed multiple disclosures to build and protect MAS Holdings' intellectual property and patent portfolio.

Assistant Manager – R&D in Flatbed Knitting Technology

TWINERY, INNOVATION BY MAS | Colombo, Sri Lanka | Full-Time December 2019 – November 2021

- Engineered functional knitted textile materials with advanced properties including moisture management, passive cooling, passive heating, and anti-bacterial treatments.
- Completed extensive technical training on Shima Seiki Whole-garment knitting, machine programming, and optimization.
- Completed specialized technical training under Prof. Tilak Dias at Nottingham Trent University (UK), focusing on Stoll and Shima Seiki machines, APEX3, and M1 Plus programming software.
- Co-developed innovative functional yarns (including fast-wicking, anti-bacterial, and fluorine-free hydrophobic yarns) in close collaboration with global yarn suppliers.
- Developed a high-performance, multilayered women's brief gusset supporting individuals with urinary incontinence and menstruation.

Executive – Smart Textiles

TWINERY, INNOVATION BY MAS | Colombo, Sri Lanka | Full-Time March 2016 – December 2019

- Engineered conductive pathways, interconnections, encapsulation, and integration methods for textile-based wearable technology products.
- Applied low-temperature plasma technology to perform hydrophobic finishes and surface activation on advanced fabrics.
- Engineered warp-knitted conductive wire solutions to support data and power transmission in smart garments.
- Developed active heating wearable platforms through the inlay of resistive heating elements in flatbed knitted structures.

EDUCATION

Postgraduate Diploma in Material Science

UNIVERSITY OF MORATUWA | Moratuwa, Sri Lanka Jan 2016 – Dec 2019

- **Research Focus:** Development of Fabric-Based Supercapacitors for wearable energy storage.

BSc. Engineering (Hons) in Textile Process Engineering

UNIVERSITY OF MORATUWA | Moratuwa, Sri Lanka Jan 2011 – Dec 2016

- **Research Focus:** Touch-sensitive wearable technology utilizing conductive fiber networks.
- **Leadership:** President (*Jan 2015 – Dec 2016*) and Vice President (*Jan 2014 – Dec 2015*) of the Textile Engineering Student Society.

KEY INNOVATION & R&D PROJECTS

- **Medical Flat Knit headgear** (A global medical technology company): Led the end-to-end textile component development utilizing flat knit technology for medical headgear and mouthpiece.
- **Vibration Therapy Glove** (A global wellness and technology company): Developed a high-performance knitted vibration glove with seamlessly integrated electronics and vibration nodes.
- **Circular Mono-Material Bra Cup** (Victoria's Secret & Pink): Successfully developed and launched a recyclable, 3D knitted mono-material bra cup solution to replace non-recyclable polyurethane cups.
- **Robotic Cover Systems** (Leading Global Robotics Brand): Developed advanced, form-fitting soft-goods cover systems for commercial robotic applications.
- **On-Demand Manufacturing business model**: Designed, developed, and deployed an agile, flatbed-knitting business model enabling on-demand customization and cost-efficient near-shoring.
- **Performance Textile Library**: Engineered diverse materials with moisture management, active/passive heating, active/passive cooling, and active compression functionalities.
- **Zonal Modulus Control**: Achieved targeted zonal modulus control by systematically varying Covered Yarn parameters (draft, TPM, and yarn counts).
- **Anti-Friction Wearable Tech** (E-sports Market): Developed a high-performance, anti-friction protective sleeve optimized for competitive gamers.

PATENTS & INTELLECTUAL PROPERTY

- **US11198961B2**: Conductive pathway (Granted)
- **EP4058624A1**: Knitted structure for an actuation element (Granted)
- **GB2605773B**: Fabric with fluid absorption capabilities (Granted)
- **US20230010185A1**: Fabric with variable fabric properties (Knitted sustainable structure for bra) (Granted)
- **US2021172101A1**: Knitted structure supporting active heating and method of manufacturing the heat-able textile (Won several awards at CES 2017) (Granted)
- **WO2022086439A1**: A 3-dimensionally-knit fabric with bi-directional variation in thickness (Published Application)
- **WO2022081094A1**: Fine gauge looking and lightweight fabric manufacturing technique through flatbed knitting machine (Published Application)
- **WO2022103334A1**: Men's boxer with 3D knitted pouch and other seamless functionalities (ISPO 2023 award-winning product) (Published Application)
- **WO2021201775A1**: Spacer fabric with moisture management (Published Application)
- **WO2022055434A1**: Knitting machine and method for knitting fabric by yarn plating with multiple yarns (Published Application)

REFEREES

Professional Referees

Contact details available upon request